

**MACHINE LEARNING-BASED STUDENT PERFORMANCE AND
DROPOUT RISK PREDICTION FOR EARLY IDENTIFICATION OF AT-
RISK STUDENTS**

PHASE I REPORT

Submitted by

Musa Haji – 230701391

Prithivi Raj – 230701508

Sachin Chaudhary – 230701402

in partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING



**RAJALAKSHMI ENGINEERING COLLEGE, CHENNAI
ANNA UNIVERSITY: CHENNAI 600 025**

September 2026

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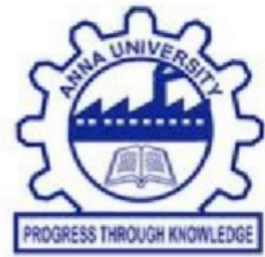
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BONAFIDE CERTIFICATE

Certified that this Report titled “**MACHINE LEARNING-BASED STUDENT PERFORMANCE AND DROPOUT RISK PREDICTION FOR EARLY IDENTIFICATION OF AT-RISK STUDENTS**” is the Bonafide work of **Musa Haji (230701391), Prithivi Raj (230701508) and Sachin Chaudhary (230701402)** who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidates.

This project addresses the following Sustainable Development Goal(s):

Goal No: 4

Description: Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.

Goal No: 10

Description: Reduce inequality within and among countries by identifying students who are at academic risk and supporting timely interventions to improve educational outcomes.

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ABSTRACT

With the growing access to student academic and behavior data, it is now possible to quickly identify which students are likely to face academic challenges. Historically, most approaches to assessing student performance involve manual grading and tracking attendance for a deep analysis of performance metrics. However, this approach does little to help uncover complex correlations between various academic, demographic, and behavioral characteristics of students. To solve this problem caused by traditional assessment methods, this research project offers to develop a machine learning system for early identification of at-risk students called Machine Learning-Based System for predicting student performance and dropouts. The system will analyze students' academic, personal, and behavioral information to categorize their performance level. The system's operational process starts with data preparation processes including, among others, filling the blanks in the data set, excluding the contradictory points, transforming categorical information, and so on. A variety of classification techniques will be covered to accomplish the task of predicting students' performance, which includes logistic regression, decision trees, random forests, k-nearest neighbors, support vector machines, and gradient boosting. Stratified cross-validation and hyperparameter optimization have been used for the purposes of having as reliable model as possible and recognizing the most suitable one among those tested. Based on the results obtained as a result of conducting experiments, Gradient Boosting turned out the most suitable model with 87.34% accuracy, 87.95% precision, 87.34% recall, and 87.42% F1-score. People, who contributed to feature importance-related analysis, found out that prior performance in school contributes significantly to the prediction made by the system and G2 grade is the most important criterion, while G1 grade, age, behavior at school, drinking habits, parents' profession, and failures also affect the outcome. Even though the present dataset does not have a particular dropout-marked objective, the created predictive framework provides a basis for the advanced dropout-risk prediction system. The coming development will involve the application of designated dropout data which will make it possible to provide accurate dropout-risk prediction.

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LIST OF ABBREVIATIONS

NLI	Natural Language Inference
PEFT	Parameter-Efficient Fine-Tuning
LoRA	Low-Rank Adaptation
RAG	Retrieval-Augmented Generation
FAISS	Facebook AI Similarity Search
ELA	Error Level Analysis
FFT	Fast Fourier Transform
PRNU	Photo-Response Non-Uniformity
XAI	Explainable Artificial Intelligence
DFD	Data Flow Diagram
DeBERTa	Decoding-enhanced BERT with Disentangled Attention
BGE	BAAI General Embedding
ViT	Vision Transformer
CNN	Convolutional Neural Network
API	Application Programming Interface

CHAPTER 1

INTRODUCTION

1.1 General

The ever-increasing access to educational data has brought with it various possible ways of examining student performance. Educational institutions are constantly gathering information regarding their students' records, attendance, demography, family background, social life, and other behavioral aspects. Manual data processing can be tiresome and rather ineffective in revealing trends that help to determine a student's performance in the future.

Student performance is influenced by numerous factors, thus it is essential to identify students in need of assistance as soon as possible for effective help. Traditional performance evaluation methods rely mainly on test scores and results of periodical tests, which may not be enough to recognize the hidden patterns in the students' performance. The use of machine learning algorithms is necessary for successful examination of historical student records. The ever-increasing access to educational data has brought with it various possible ways of examining student performance. Educational institutions are constantly gathering information regarding their students' records, attendance, demography, family background, social life, and other behavioral aspects. Manual data processing can be tiresome and rather ineffective in revealing trends that help to determine a student's performance in the future.

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1.2 Objective

The primary objective of the system being proposed is to implement a machine learning-enabled system for predicting students' performance and possible dropout risks, which may aid in understanding and analyzing all student-related academic, demographic, social and behavioral aspects.

The specific objectives are:

1. To gather student academic data, social data, demographic data, and behavioral data to analyse it.
2. To process the student data collected to deal with missing values, records that are inconsistent and duplication of data.
3. To encode categorical attributes and convert numerical data into a form that can be used in machine learning.
4. To create training and testing datasets out of the data processed so it can be utilized in model development and evaluation.
5. To determine the factors that may affect the academic performance of students.
6. To implement machine learning classification algorithms.
7. To assess the model performance through Accuracy, Precision, Recall, and F1 score evaluation.
8. To perform stratified cross-validation to test how reliable and consistent the produced models are.
9. To execute hyper parameter tuning for making machine learning models more efficacious.
10. To categorise students into the Low, Average, and Good performance groups.
11. To calculate the probability of prediction regarding every particular student.
12. To employ the analysis of features that influence the predictions of student performance.

To provide an early identification mechanism for students who may require additional academic guidance and intervention.

1. To establish a foundation for extending the system with a dedicated dropout-risk prediction module using an appropriately labelled dropout dataset.

Implementation Objective

For the present project implementation, the scope is restricted to:

- Student Data Collection and Preprocessing
- Categorical Encoding and Numerical Feature Transformation
- Student Performance Classification
- Machine Learning Model Training and Comparison
- Cross-Validation and Hyperparameter Tuning
- Performance Evaluation using Accuracy, Precision, Recall, and F1-score
- Feature-Importance Analysis
- Individual Student Performance Prediction
- Prediction Probability Generation
- Early Identification of Students Requiring Academic Support

1.3 Existing System

Multiple approaches and machine learning models have been developed for analysing and predicting student academic performance. Traditional educational systems generally depend on examination marks, attendance records, assignments and periodic assessments to evaluate students. Statistical techniques and conventional classification algorithms are also used to analyse student-related data and identify factors that influence academic outcomes. Datasets such as the Student Performance dataset provide academic, demographic, social and behavioural attributes that can be used for predicting student performance.

Apart from traditional statistical approaches, various machine learning algorithms such as Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors, Support Vector

Machine and Gradient Boosting have been applied to student performance prediction. These models can identify patterns within historical student records and classify students according to their expected academic performance. However, the effectiveness of these approaches depends heavily on the quality of the dataset, appropriate preprocessing, feature selection and the distribution of the target classes. In cases where some performance categories contain very few students, class imbalance can negatively affect the prediction results.

The major limitation of existing student performance assessment systems is that they primarily focus on analysing academic results after the completion of assessments rather than providing an early indication of students who may require support. Manual analysis of multiple academic, demographic and behavioural factors can also be time-consuming and may overlook relationships between different attributes. Furthermore, many prediction systems provide only the final classification without clearly indicating which factors contributed to the prediction. Therefore, a system that combines machine learning-based prediction with model evaluation and feature-importance analysis is highly useful for supporting early identification of students who may need academic intervention.

1.4 Proposed System

The proposed system is a machine learning-based student performance and dropout risk prediction framework.

The system accepts student academic, demographic, attendance, behavioural, social, and family-related data and performs the following high-level operations:

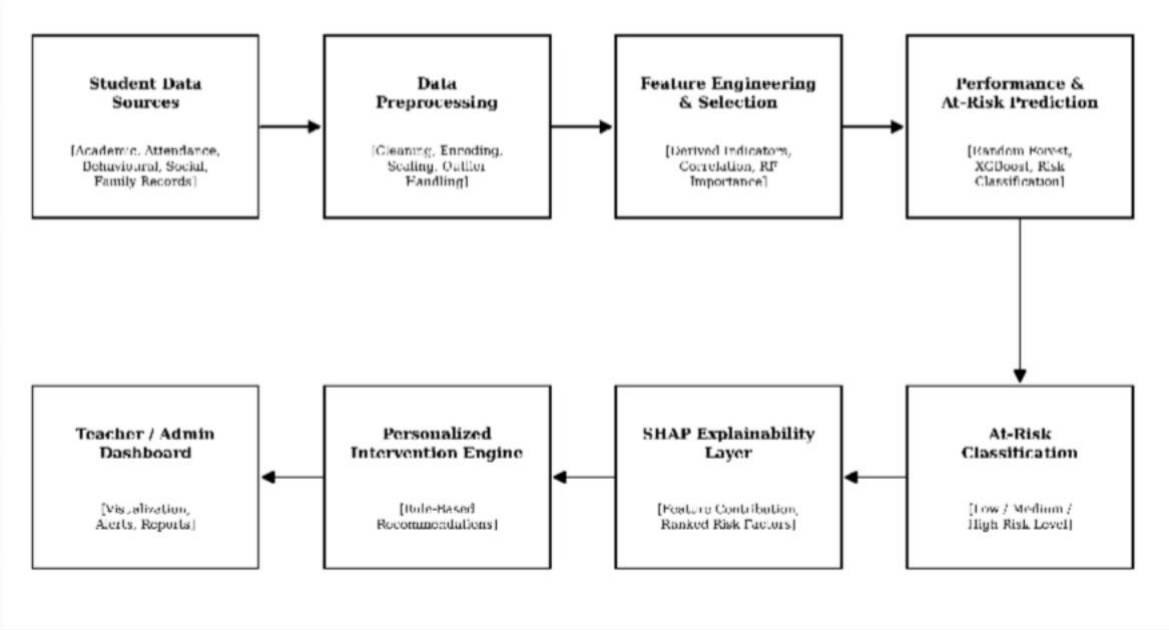


Figure 1.4: System Architecture

The system consists of the following major conceptual components:

1. Student Data Collection
2. Data Preprocessing
3. Feature Engineering and Selection
4. Performance Prediction
5. Risk Classification
6. Explainability Layer
7. Intervention Support
8. Teacher/Administrator Dashboard

The current implementation concentrates on **Data Preprocessing, Performance Prediction, Model Evaluation, and Feature Importance Analysis.**

Data Preprocessing

The module:

- cleans student data,
- handles missing values,
- encodes categorical features,
- prepares numerical features,
- divides the dataset for training and testing.

Performance Prediction

The prediction module:

- trains different machine learning models,
- compares their performance,
- performs cross-validation and tuning,
- predicts student performance as **Low, Average, or Good.**

Explainability

The explainability layer identifies the important factors contributing to a student's prediction and helps provide an understandable basis for the model's output.

Risk Identification

The system uses student performance and relevant factors to identify students who may require additional academic attention. The framework can later be extended to direct **Low / Medium / High dropout-risk prediction** using a suitable dropout-labelled dataset.

CHAPTER 2

LITERATURE SURVEY

2.1 Literature Review

The growing popularity of the utilization of educational data and machine learning approaches has led to a substantial amount of interest regarding predicting the academic performance of students and identifying those students who need academic support from an early age. Prior studies have utilized a variety of factors including academic, demographic, behavioral, and social for predicting student outcomes. This literature review highlights the important publications regarding student performance prediction, classification using machine learning, features, and early at-risk identification.

2.1.1 Student Performance Datasets and Academic Prediction

Performance prediction of the student was done by Cortez and Silva (2008) based on the information obtained from secondary schools. The study focused on the application of machine learning approaches to make predictions about the grades of students and also identified factors linked with academic performance of students.

The UCI Student Performance Database is another dataset which has been extensively used for the purpose of making predictions about the performance of the students. The database contains all the information about the academic performance of students, their family backgrounds, social activities, study habits and many other behavioral factors.

Machine learning methods that have been used for student performance classification include Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, K-Nearest Neighbors, and Gradient Boosting. This helps to recognize the patterns in the student data and classify the students based on their potential academic performance.

Machine learning techniques like Random Forest and Gradient Boosting are more suitable as they take into account the information provided by several different features as well as the non-linear relationship in the data set. But the prediction accuracy is influenced by class imbalance, quality of features and suitable model parameters.

2.1.2 Feature Importance and Explainable Prediction

Academic predictive models must not only give a prediction result but also help educators to better understand what affects the given prediction result. Feature-importance

methods can be used to determine which attributes affect the model the most. Explainable AI approaches like SHAP (SHapley Additive exPlanations) allow interpreting a prediction result in terms of contributions made by particular features. Such methods may enhance the transparency of machine learning-driven education-related models and help to get more useful predictions from them.

Identifying students who might suffer poor academic results early on is one of the key applications of educational data mining. Based on the analysis of past academic performance and student attributes, prediction methods can help detect students who will need extra assistance.

But most of the existing methodologies concentrate mostly on predicting grades or overall academic performance rather than developing an early warning system that includes all the steps necessary. Moreover, predicting dropouts directly implies the availability of a dataset with labelled dropout records.

2.2 Comparison and Discussion

Study/System	Performance ML Prediction	Feature Analysis	Early Identification	Explainability
Cortez & Silva – UCI Student Performance Dataset	✓	Limited	Limited	✗
Traditional ML Approaches	✓	Limited	✗	✗
SHAP-based Approaches	✓	✓	Limited	✓
Proposed System	✓	✓	✓	✓

Table 2.2: Comparison of Existing Systems

2.3 Conclusion

Literature shows that machine learning tools could be efficiently applied for student data analysis and academic performance prediction. Current research indicates the significance of academic, demographic, social, and behavioral features of students and the influence of these features on the performance of students. Different classification models have also been used for prediction of student performance in current researches, which makes a good basis for development of predictive educational systems.

However, current systems pay much attention to prediction and offer little information on the factors that determine the predicted result. In addition, there is a need for systems that will be able to find students in need of academic assistance before examinations, not after their results. Explainable machine learning methods will add additional value to such systems due to identification of factors influencing the prediction.

The comparison and evaluation of models are essential steps for the selection of a reliable prediction mechanism. Various methods such as cross-validation and hyperparameter optimization will facilitate the estimation of consistency in the models and the improvement of the predictions' performance. Moreover, the extraction of important features will facilitate the understanding of factors that influence student results more.

In conclusion, the proposed solution combines data pre-processing, prediction of student performance through machine learning techniques, evaluation of the model, feature importance estimation, and detection of students in need of academic assistance. The solution serves as a basis for data-driven educational interventions and can contribute to making decisions about student assistance. In addition, the proposed solution can be expanded by predicting the risk of student dropout through an appropriate labelled dataset.

CHAPTER 3

SYSTEM DESIGN

3.1 System Architecture

Figure 3.1 presents the architecture of the Machine Learning-Based Student Performance and Dropout Risk Prediction System. The process starts from the data source that consists of data about academics, attendance, behaviour, social and family details of students. The Data Preprocessing Module will clean and scale the data along with feature engineering and selection for determining the right features.

The selected features are fed into the Performance and At-Risk Prediction module which employs machine learning models like Random Forest and XGBoost. The next stage will be At-Risk Classification to determine the risk of the particular student. In the next layer SHAP Explainability Layer determines the important predictors and in the Personalized Intervention Engine, the right suggestions are provided. The Teacher/Administrator Dashboard will provide visual representations to the users.

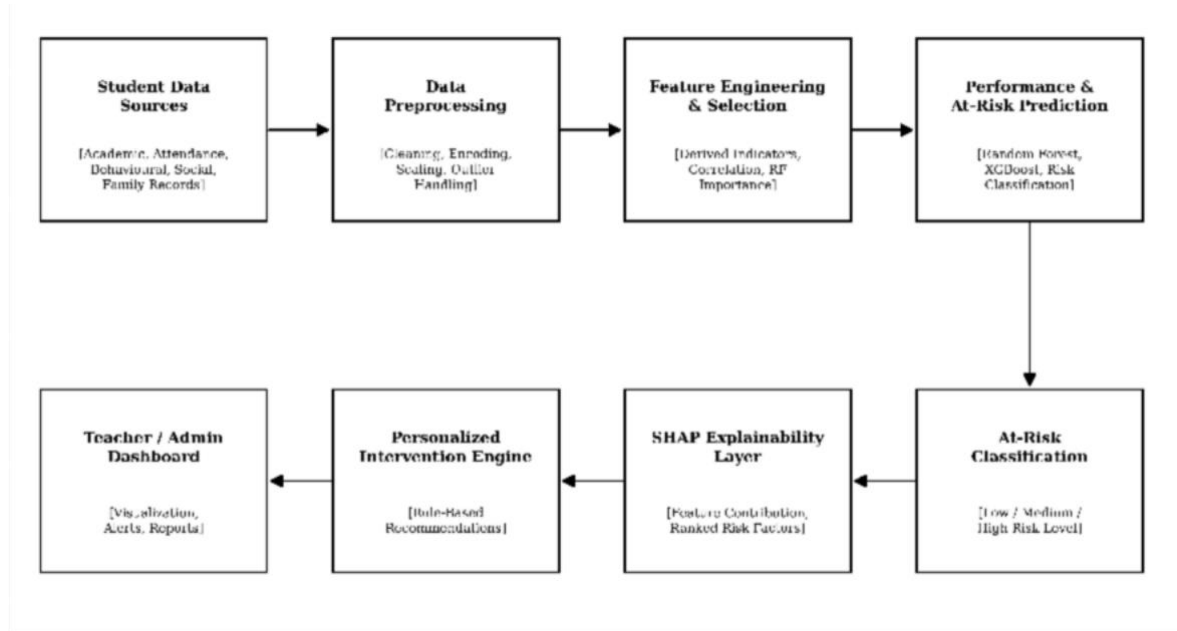


Figure 3.1: Overall architecture of the proposed machine learning-based student performance and dropout risk prediction system.

The first component of the architecture is the Student Data Sources component. This component serves as an input layer of the architecture and supplies the necessary information for predictions. Such information may comprise academic data, attendance data, behavioral attributes, social attributes, and family attributes. Different attributes enable the architecture to take into consideration several aspects of a student's profile and academic performance instead of relying solely on the exam scores.

The collected information is supplied further to the Data Preprocessing component. The raw data about the students may have missing values, categorical attributes, inconsistencies, and attributes with different scales. In order to apply machine learning methods to the data, preprocessing is carried out first. The processes involved in this process include data cleaning, categorical attributes transformation, scaling, and other necessary data processing.

Next, the data goes to the Feature Engineering and Selection module. At this step, the goal is the discovery of features that will help predict student performance. The derived indicators can be calculated based on available features, and the contribution of each can be studied using correlation analysis or by evaluating feature importance from the models. Feature selection allows the prediction step to consider only valuable information about students and minimize the impact of irrelevant one.

The resulting features are sent to the Performance and At-Risk Prediction module. This is the machine learning step of the architecture. Models are built using the processed dataset of students, and learn the relationship between inputs and outputs. Models like Random Forest or XGBoost can be employed for prediction at this step of the proposed architecture. At the end, the result of the prediction stage is the performance output as well as additional information needed to find at-risk students.

The output of the prediction goes through the At-Risk Classification module. This is a phase where the output of the model is transformed into risk categories. The architecture of the model employs the use of risk levels known as Low, Medium, and High Risk Level. It makes the prediction more interpretable and enables the educators to prioritize their students who need additional attention.

This leads us to the SHAP Explainability Layer. Just the fact that the prediction was made does not really show why a certain student received the said prediction. The explanation comes from the analysis of the contribution of each individual feature to the model prediction. SHAP-based feature contribution helps in identifying and ranking the features which contribute greatly to the prediction of a certain student.

Information that is generated in the processes of prediction and explainability is provided to the Personalized Intervention Engine. This engine utilizes the risk factors and predictions generated from the previous stages to formulate rules. For instance, students in need of academic intervention can be identified using the data collected. The objective of this process is to derive intervention information from the results of predictions and not only classify the problem.

The last process involves the presentation of the results in the Teacher/Administrator Dashboard. The dashboard is the output and monitoring part of the framework. This means that the dashboard will provide information on the performance of students in terms of risk and visualization.

In summary, the architectural design provides for a full flow of student data acquisition → data preprocessing → feature extraction → prediction through machine learning → risk categorization → explanations → interventions → dashboard visualizations. The current design mainly focuses on the establishment of the data processing and student performance prediction process, but the architecture provides the basis for expanding the process to predict dropouts using appropriate labeled data.

3.2 System Requirements

These sections cover the Software Requirements, Hardware Requirements, Dataset Requirements, and Deployment Requirements required to build the proposed Machine Learning Based Student Performance and Dropout Risk Prediction System.

3.2.1 Software Requirements

Component	Requirement
Operating System	Windows/Linux
Programming Language	Python
Development Environment	Jupyter Notebook / Google Colab
ML Framework	Scikit-learn

Visualization	Matplotlib, Seaborn
Explainable AI	SHAP
Data Storage	CSV / Pandas DataFrame
Input Format	CSV
Output Format	Prediction Report / DataFrame
Prediction Output	Low / Average / Good
Prediction Probability	Class Probability Scores

Table 3.2.1: Software Requirements

SHAP is a part of the explainability module because it allows us to determine the role of each particular feature of the students in the outcome of the prediction by the model. We will be able to see the role played by such characteristics as grade, age, absence, and failures in the prediction of academic performance.

3.2.2 Hardware Requirements

Provisioning of the hardware resources is necessary in order to ensure that the development, testing, and application of the student performance predictor works smoothly. The machine learning pipeline that was designed includes the following steps: data preprocessing, feature transformation, model training, cross-validation, hyperparameter tuning, and prediction, which all require sufficient CPU and memory. As the machine learning models were constructed using traditional machine learning methods, there is no need to have a dedicated GPU. Table 1 shows the hardware configuration that is necessary to run the system. The minimum configuration is sufficient for development, data preprocessing, model training, debugging, and testing, whereas the recommended one allows additional computation and memory capacity for further model training, cross-validation, and larger data sets.

Component	Minimum Configuration	Recommended Configuration
Processor (CPU)	Intel Core i3 / Equivalent Multi-Core CPU	Intel Core i5 or above
System Memory (RAM)	4 GB	8 GB or above
Graphics (GPU)	Not Required	Optional NVIDIA GPU
Storage (SSD)	256 GB SSD	512 GB or above SSD

Table 3.2.2: Hardware Requirements

Dataset Requirements:

Student Performance Dataset

Dataset includes:

- Student Performance Dataset — academic and demographic information
- 395 student records
- 33 original features
- 32 input features after target removal
- 58 processed features after encoding and transformation
- 316 training records
- 79 testing records
- Low, Average, Good — performance classes

The Student Performance Dataset is used to develop and evaluate the machine learning-based student performance prediction system. It contains academic, demographic, social, and behavioural attributes of students that can be used to identify factors associated with academic performance. During preprocessing, missing and inconsistent values are handled,

The Student Performance Dataset is used to design and test the machine learning-based student performance prediction system. It consists of various academic, demographic, social, and behavioral characteristics of students which can help in identifying the factors that affect the academic performance of the students. While preprocessing, the missing and inconsistent values are handled,

the categorical attributes are encoded, and the input attributes are prepared in the desired format for machine learning modeling. The four classes in the original target attribute i.e., Low, Average, Good, and Excellent have been reduced to three classes due to the presence of only 19 records belonging to the Excellent category which creates severe class imbalance. The three target classes are now Low, Average, and Good. There are 316 training records and 79 testing records in the current dataset. The machine learning models are tested on accuracy, precision, recall, and F1-Score. The current dataset is used for predicting student performance; however, a new dropout dataset is needed for predicting student dropouts.

3.3 Activity Diagram

Multimodal veracity verification is done by separating the input into two streams, the first for the text which is analyzed based on fact checking through NLI, and the second stream for the image, which is processed using spatial and frequency domain forensics, and the output from these two processes is integrated together using dynamic weighting and contextual reasoning in order to arrive at the veracity conclusion. Input is separated into two streams: the text is processed through claim extraction, evidence retrieval, re-ranking and NLI verification; the media content is analyzed using spatial, frequency and map fusion forensics as shown below.

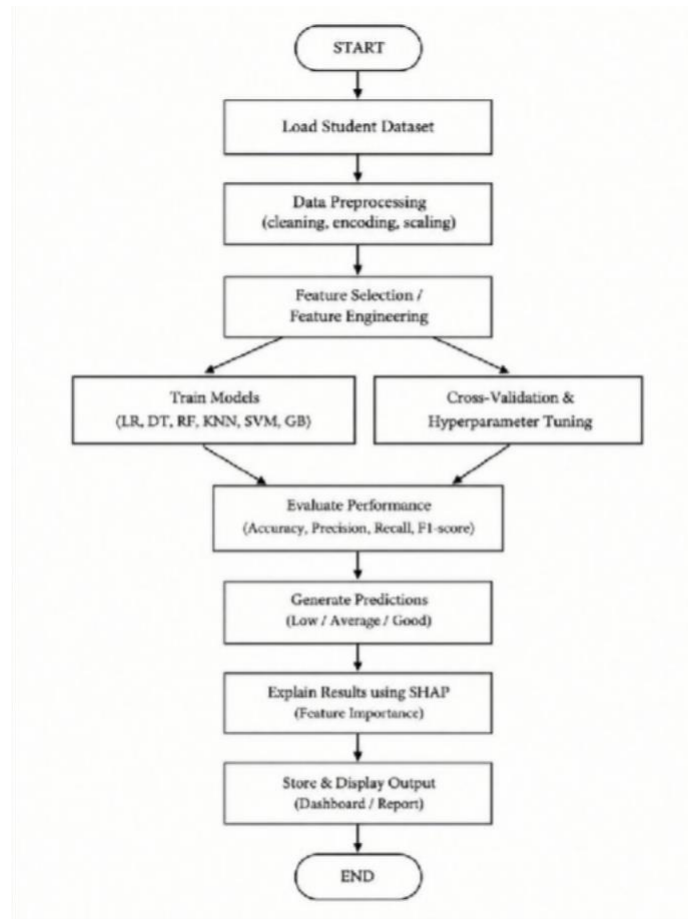


Figure 3.3: Activity Diagram

3.4 Data Flow Diagram

The Data Flow Diagram depicts the overall process flow of the student performance prediction system from the student dataset and data collection to data processing, after which data goes through feature selection and training of the model. At the same time, cross-validation and hyperparameter optimization are done to optimize and assess the models. Feature analysis using SHAP is applied to the processed features to detect the most impactful factors on the predictions. As a result, the model predicts the student's performance as Low, Average, or Good with probabilities and gives the final result.

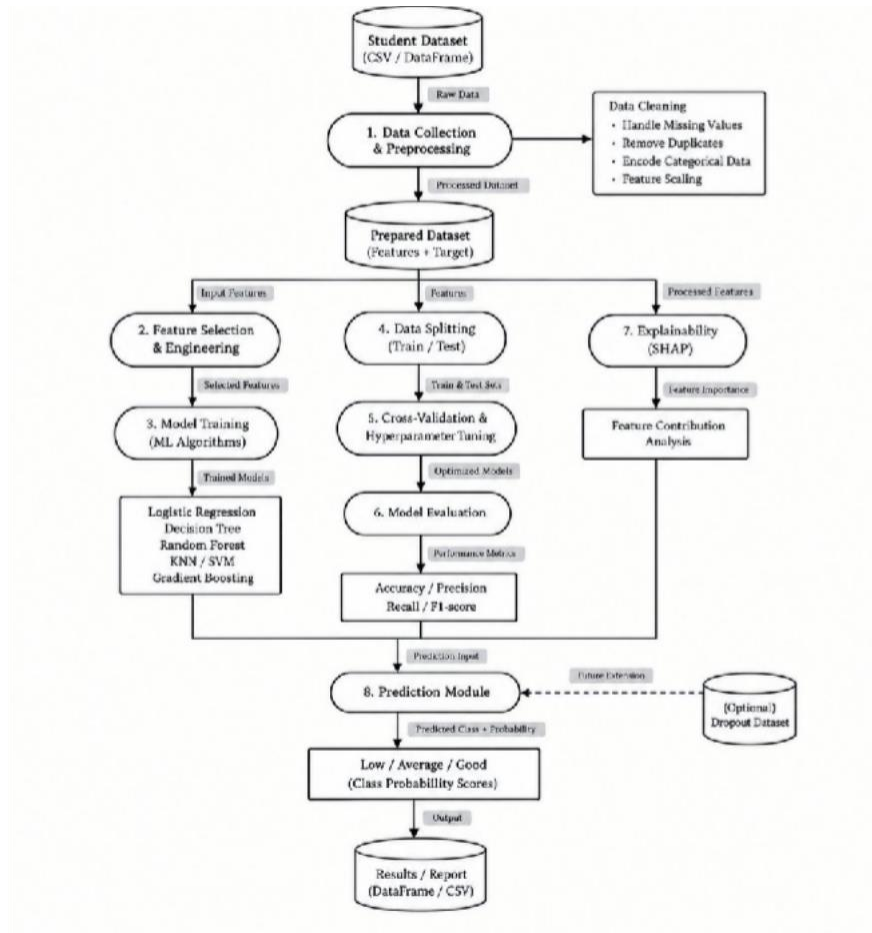


Figure 3.4: Data Flow Diagram

CHAPTER 4

PROJECT DESCRIPTION

4.1 -Methodologies

The suggested methodology is based on a modular prediction approach that utilizes machine learning algorithms. Instead of employing a single classification method for predicting students' performance, it instead breaks the whole process down into several steps, comprising data preprocessing, feature generation, the training of the prediction model, evaluation of the model, prediction, and explanation..

4.1.1 Machine Learning-Based Performance Prediction

The prediction module performs the entire process of training and evaluating machine learning models from the preparation of student data to the generation of the final performance category. Different classification algorithms are utilized and assessed to find the best model for predicting the academic results of students. Thus, this provides a structured evaluation of various models and allows for reliable identification of students with academic difficulties.

4.1.2 Data Preprocessing and Feature Engineering

The processing of student information takes place prior to its consumption by forecasting models. The data processing stage deals with the absence of data, eliminates duplicate and/or inconsistent information, transforms the values, and applies the appropriate rules for encoding categorical variables.

The processing proceeds in the following order:

Stage 1: Data Cleaning

The preliminary student dataset is cleansed by dealing with missing data, removing duplicates, and making the data consistent. This makes sure that the data fed into the machine learning model is appropriate for analysis.

Stage 2: Feature Transformation

Categorical features are transformed into numbers, and then these numbers undergo scaling. After that, the next steps involve carrying out feature generation and selection to prepare a

relevant feature set for building the model.

Stage 3: Model Training and Evaluation

The dataset that has been processed has been split into a training set and a test set. Models that are used include Decision Trees, Logistic Regression, Random Forests, KNN, SVMs, and Gradient Boosting, which are trained and evaluated on the basis of their accuracy, precision, recall, and F1 Score. Cross-validation and hyperparameter tuning were used to improve the performance of the models.

Stage 4: Explainable Prediction

The concept of SHAP is utilized in understanding the influence of individuals' features on its output. As a result, this analysis allows one to detect relevant parameters impacting the students' performance. The result of the system denotes whether the category predicted is Low, Average, or Good and shows the respective probabilities.

4.2 Modules

4.2.1 Data Collection and Preprocessing Module

The Data Collection and Preprocessing Module is responsible for receiving the student performance dataset and preparing it for machine learning.

Functions

- Dataset loading
- Missing value handling
- Duplicate removal
- Inconsistent data handling
- Categorical data encoding
- Numerical feature transformation
- Feature scaling
- Train-test data preparation

The module takes unprocessed records of students as input which it processes into a formatted dataset that can be used for machine learning. The dataset comprises of student characteristics related to academics, demography, sociology and behavior and goes through scrubbing before proceeding to further processing.

Output

Processed Student Dataset



Cleaned Features + Target

4.2.2 Feature Selection and Engineering Module

The content of this module meets the requirements for the most important predictors of student academic outcomes. Categorical attributes undergo transformation to be given their numerical forms after preprocessing.

A. Feature Transformation

Categorical variables are encoded into numerical values so that they can be processed by machine learning algorithms. Numerical attributes are also transformed and scaled where required.

B. Feature Selection

The module finds suitable student characteristics that affect academic performance. Specific aspects taken into account in the model include the past grades, age, attendances, previous failures and behavioural traits.

The dataset that has been processed has 58 components after the encoding and transformation operations are complete.

4.2.3 Machine Learning Model Training

The Machine Learning Model Training Module is in charge of training a number of classification methodologies utilizing the created dataset of students.

The models that are trained include:

- Logistic Regression
- Decision Trees
- Random Forests
- K-Nearest Neighbors
- Support Vector Machines
- Gradient Boosting

The dataset will be split into 316 pieces that will be used for training and 79 records meant for testing. All models are trained on training data and later tested with testing data.

The target variable is classified into three performance categories:

Low → Average → Good

Initially, the four-class conceptualization featured an Excellent category, which encompassed just 19 students, causing a severe case of class imbalance. Hence, a three-class conceptualization was utilized for educational purposes.

4.2.4 Cross-Validation and Hyperparameter Tuning

The The Cross-Validation and Hyperparameter Tuning Module enhances the efficiency and reliability of learning models.

A.Stratified Cross-Validation

Performing five-fold stratified cross-validation allows to maintain the percentages of performance classes for various folds. Such approach makes it possible to obtain more accurate predictions than using only a single train-test split.

B.Hyperparameter Tuning

Grid Search is carried out for appropriate machine learning algorithms to find the necessary parameters. The algorithms used for tuning by cross-validation are Random Forest and Gradient Boosting.

The obtained highest F1-scores in cross-validation were 0.8700 for Random Forest, 0.8803 for Decision Tree, and 0.8891 for Gradient Boosting.

4.2.5 Model Evaluation Module

The Model Evaluation Module compares the performance of the trained classification models using standard evaluation metrics.

The evaluation metrics include:

- Accuracy
- Precision
- Recall
- F1-score

The test dataset is used to evaluate the models and their capacity to classify student performance into categories. The final model selected is the tuned Gradient Boosting model, which has reached an Accuracy of 87.34%, Precision of 87.95%, Recall of 87.34%, and F1-score of 87.42%. This evaluation gives grounds to select the model which is best suited for the final prediction module.

4.2.6 Student Performance Prediction Module

Using the chosen machine learning model, the Student Performance Prediction Module makes the predictions of the created model on the newly entered student records.

The input student features are processed in the same manner as they were prepared for model training: using the preprocessing and feature transformation methods.

After that, the trained Gradient Boosted model is utilized to predict in which academic performance category the student would fall.

The output consists of:

Student Input



Preprocessing



Trained Model



Performance Prediction



Low / Average / Good

4.2.7 Explainable AI Module

The Explainable AI Module serves to clarify what elements impact machine learning model outcomes. For that purpose, SHAP (SHapley Additive exPlanations) method is used to analyze feature importance and make predictions more interpretable.

This module presents the features that most affect student performance. G2 is shown to be the leading factor in the model created, along with the following features, G1, Age, Going Out Frequency, alcohol consumption, parental occupation, Absences and Past Failures.

This module allows making the prediction process more transparent as it elucidates the reasons behind a prediction instead of just providing the final classification.

CHAPTER 5

IMPLEMENTATION AND RESULT DISCUSSION

5.1 Implementation and Results

The present document describes how the AI-Based Student Academic Performance and Risk Prediction System has been implemented. Key features of the developed system include student data management, academic performance forecasting, risk analysis, comparison of predictions as well as providing an explanation for predictions. Implementation allows an interactive interface which enables the administrator to access and evaluate information about students, conduct predictions and carry out model analysis of student performance.

5.1.1 Data Pre-processing

The initial step in the pipeline implementation involves the conversion of the raw student dataset to a structured and clean format that is appropriate for machine learning.

Operations:

The system carries out preprocessing operations as follows:

Handling Missing Values: Identification and processing of missing values in order to keep consistency in the dataset.

Removal of Duplicates: Elimination of duplicates and inconsistent records to eliminate unnecessary bias in the process of model training.

Encoding of Categorical Features: Transformation of categorical attributes of students into a numerical form suitable for machine learning algorithms.

Transformation of Features: Transformation of input features into a unified numerical format.

Scaling of Features: Scaling of numerical features wherever needed so that they will serve as input data for KNN, SVM, and Logistic Regression algorithms.

After preprocessing and encoding, 33 original features transform into 58 processed features that are used for model training.

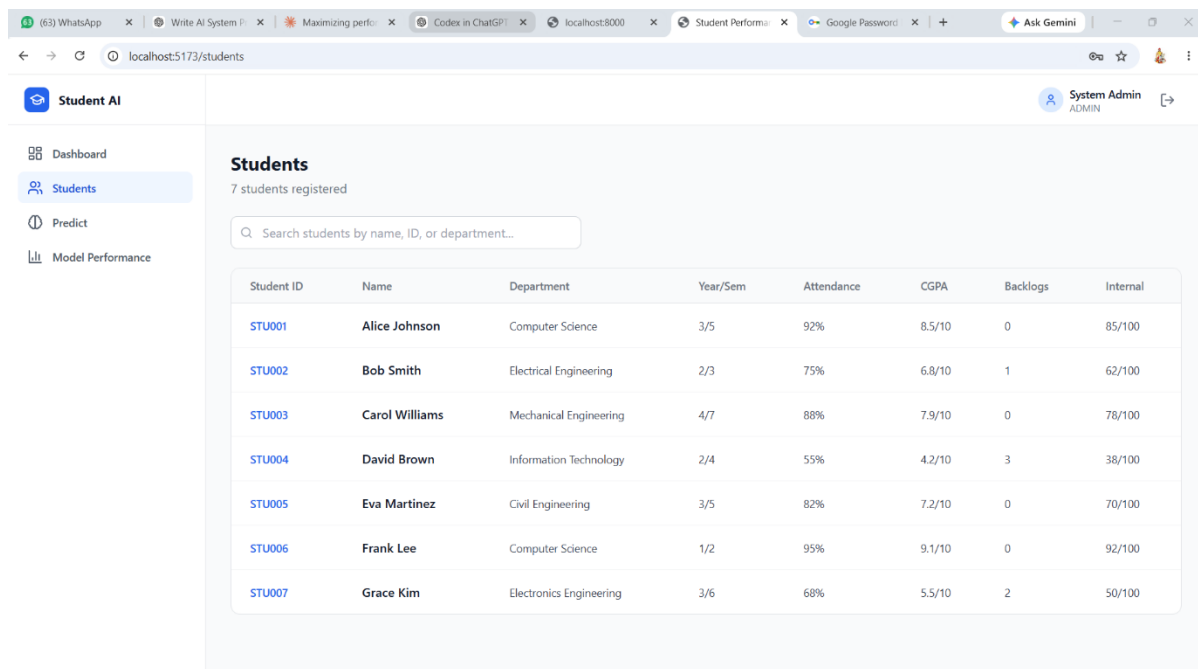
5.1.2 Dataset Preparation and Student Management Interface

The Student Performance Dataset contains **395 student records** with academic, demographic, social, and behavioural attributes. The processed dataset is divided into training and testing subsets for model development and evaluation.

Dataset Component	Records / Features
Total Records	395
Original Features	33
Processed Features	58
Training Records	316
Testing Records	79

Table 5.1.2: Dataset Preparation

The product also includes a Student Management Interface that lets you look through and handle student records. In the interface shown, the admin can learn some details such as Student ID, Name, Department, Year/Semester, Attendance, CGPA, Backlogs, and internal marks. A search tool has also been introduced that permits to search for students by name, ID, and department.



The screenshot displays a web application titled "Student AI" with a sidebar menu containing "Dashboard", "Students", "Predict", and "Model Performance". The "Students" section is active, showing "7 students registered". A search bar is present with the placeholder text "Search students by name, ID, or department...". Below the search bar is a table with the following data:

Student ID	Name	Department	Year/Sem	Attendance	CGPA	Backlogs	Internal
STU001	Alice Johnson	Computer Science	3/5	92%	8.5/10	0	85/100
STU002	Bob Smith	Electrical Engineering	2/3	75%	6.8/10	1	62/100
STU003	Carol Williams	Mechanical Engineering	4/7	88%	7.9/10	0	78/100
STU004	David Brown	Information Technology	2/4	55%	4.2/10	3	38/100
STU005	Eva Martinez	Civil Engineering	3/5	82%	7.2/10	0	70/100
STU006	Frank Lee	Computer Science	1/2	95%	9.1/10	0	92/100
STU007	Grace Kim	Electronics Engineering	3/6	68%	5.5/10	2	50/100

Figure 5.1.2: Student Management Interface

The student management element serves as the main source and supervision configuration for the system. It enables the categorization of student records and allows for prior review before making predictions. The information shown gives an organized form of parameters regarding academics that can thus be processed further by use of the system in the future.

5.1.3 Machine Learning Model Results

The system also assesses various machine learning classification algorithms to determine which one is the most appropriate for predicting student academic performance. The models being utilized include:

- Logistic Regression
- Decision Tree
- Random Forest
- K-Nearest Neighbor
- Support Vector Machine
- Gradient Boosting

Model Performance

Model	Accuracy	F1-Score
Decision Tree	87.34%	87.31%
Logistic Regression	86.08%	86.19%
Random Forest	86.08%	86.16%
Gradient Boosting	84.81%	84.89%
SVM	78.48%	78.36%
KNN	70.89%	71.07%

Table 5.1.3: Machine Learning Model Performance

The first results display that Decision Tree delivered the best test accuracy amongst the initially evaluated untuned algorithms at the same time that the rest of the algorithms showed varying levels of performance in the different propensity classes of students.

5.1.4 Cross-Validation Results

To assess the consistency of the models across different subsets of the dataset, 5-fold stratified cross-validation was performed.

Model	Mean F1-Score	Standard Deviation
Random Forest	0.8534	0.0653
Gradient Boosting	0.8509	0.0297
Decision Tree	0.8342	0.0237
Logistic Regression	0.8312	0.0486
SVM	0.7979	0.0306

Model	Mean F1-Score	Standard Deviation
KNN	0.6637	0.0394

Table 5.1.4: 5-Fold Stratified Cross-Validation Results

Results from the cross-validation indicate that Random Forest has the highest mean F1-score among the different models that were previously examined, and Gradient Boosting also performed well but less variably across folds.

5.1.5 Hyperparameter Tuning and Final Model Evaluation

Grid Search was applied to selected machine learning models to identify suitable hyperparameter combinations and improve their predictive performance.

Model	Best Cross-Validation F1-Score
Random Forest	0.8700
Decision Tree	0.8803
Gradient Boosting	0.8891

Table 5.1.5: Hyperparameter Tuning Results

The optimized Gradient Boosting model reached the top cross-validation F1 score of 0.8891 and was chosen as the final academic performance forecasting model.

The chosen model was subsequently tested on the testing set.

Metric	Result
Accuracy	87.34%
Precision	87.95%
Recall	87.34%
F1-Score	87.42%

Table 5.1.5: Final Model Evaluation Results

The final model has a performance score of 87.34%, while the F1-score of the model is equal to 87.42%, indicating that it can classify students according to the performance categories.

5.1.6 Classification Dashboard and Risk Assessment Analysis

The developed system has an administrative dashboard, which provides consolidated information about students' performance and other prediction-related information.

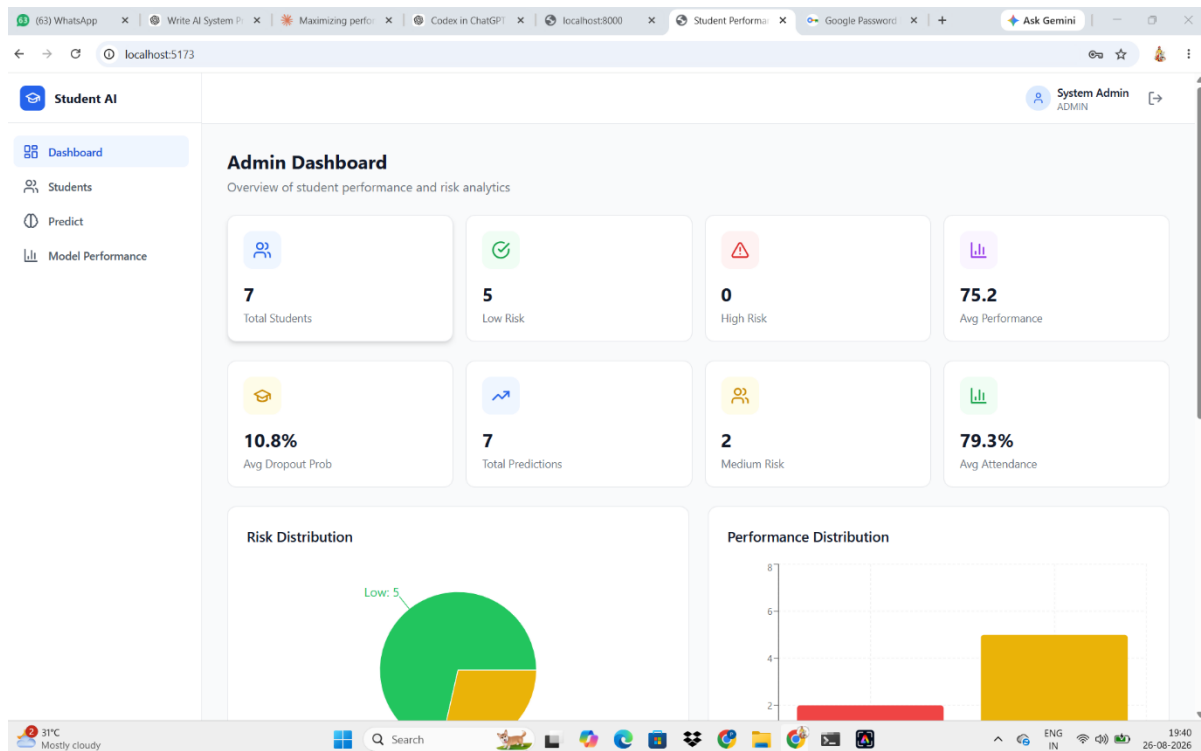


Figure 5.1.6: Administrative Dashboard and Risk Analysis

The dashboard shows key indicators at the system level, such as the sum total of registered students, the number of students categorized as low-risk, medium-risk, and high-risk respectively; the average performance in studies as well as the average dropout probability. The practical example seen on the screen shows that the dashboard has 7 students in total, out of which 5 belong to the Low Risk category and 2 belong to the Medium Risk one. Only a number of students have been identified as belonging to the High Risk category.

Moreover, the visual analytic area has two types of charts: Risk Distribution and Performance Distribution. These charts make it possible to summarize the outcomes of predictions using graphical data representation.

5.1.7 Recent Predictions and Prediction History

The user interface for prediction history provides information about previous forecasting results.

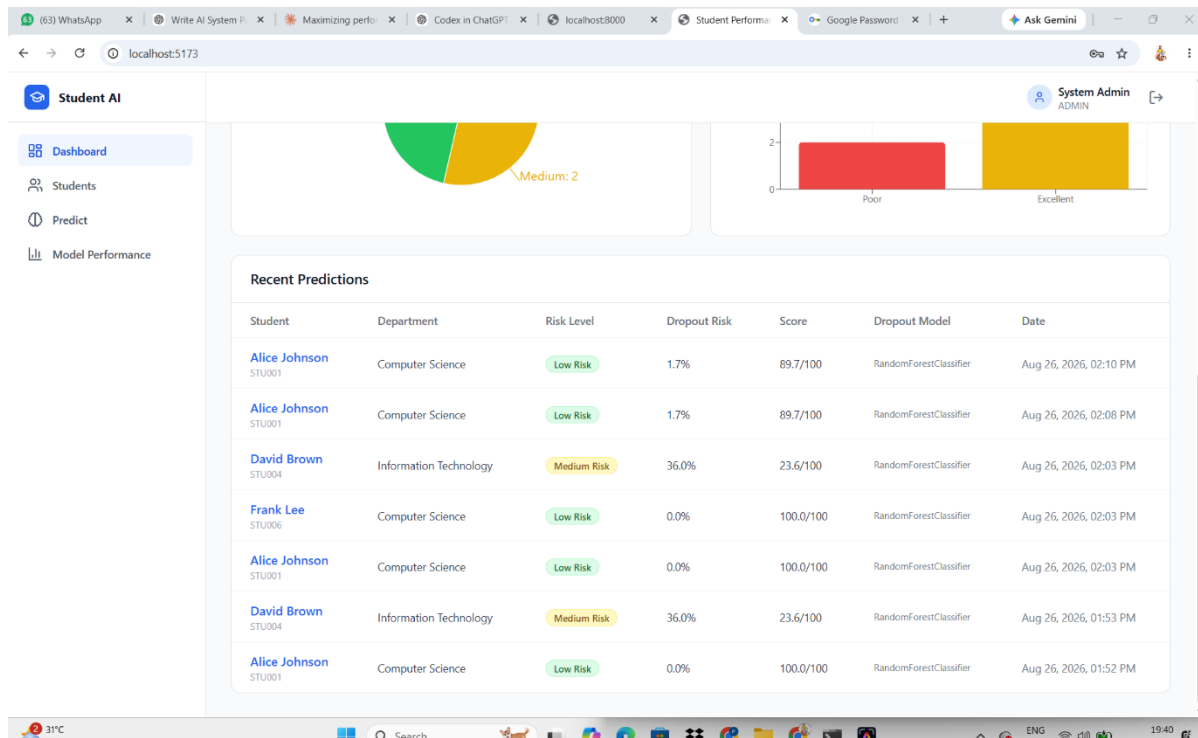


Figure 5.1.7: Recent Student Predictions

The information is displayed in a tabular format and has the following fields: student name, ID, department, risk level, dropout probability, academic performance score, model, and date & time of the prediction. The picture from the user interface can be used to track single predictions after the model has been executed. For instance, the user interface labels students with names such as “Low risk” or “Medium risk” and provides corresponding numerical values of dropout risk. This feature provides an opportunity to keep a history of the prediction and allows users to compare results of students for different time frames.

Important: Note that your current dataset does not have a labeled dropout target, therefore, you should be very careful in your report. You may define this interface as a risk monitoring/output component that was implemented, but do not state that the dropout percentages shown are the results of the validated dropout model, unless you have trained the model in a separate way and validated it.

5.1.8 Model Performance Comparison

The implementation includes a dedicated **Model Performance Interface** for comparing the machine learning algorithms used within the system.

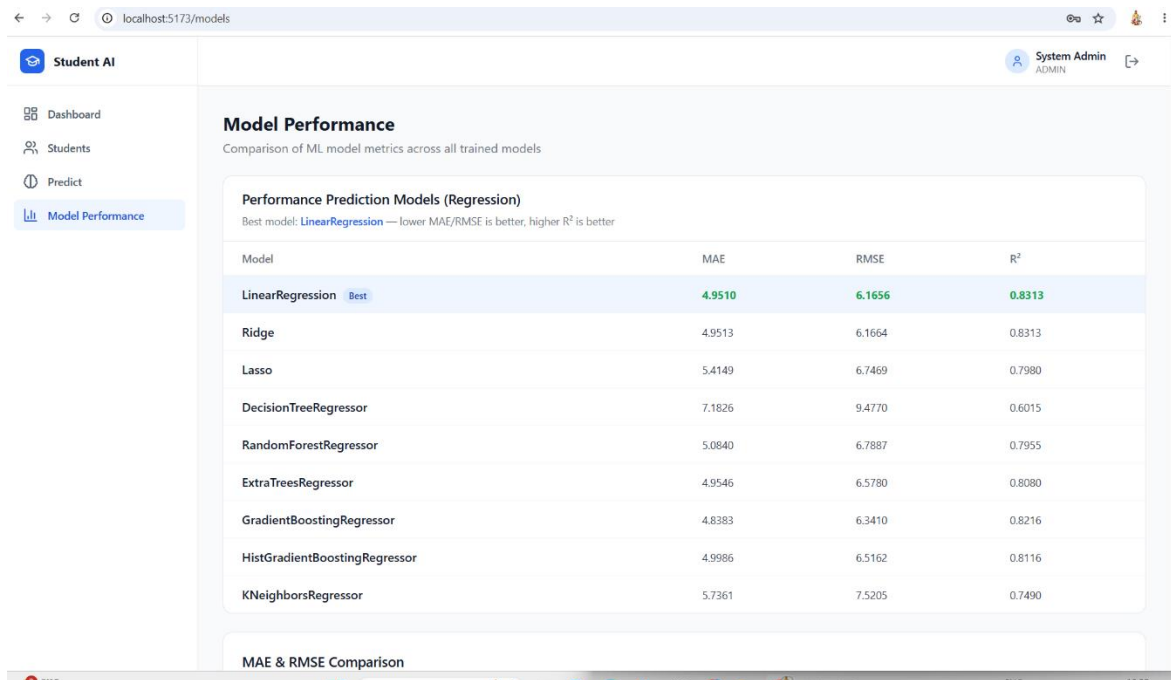


Figure 5.1.8: Machine Learning Model Performance Comparison

The interface shows the performance of many classification models with the help of tables and graphs. The models that are showcased include algorithms like Logistic Regression, Decision Tree, Random Forest, Extra Trees, Gradient Boosting, AdaBoost, K-Nearest Neighbors, Support Vector Classifier, and Histogram Gradient Boosting in accordance with a comparison configuration that has been selected by the user.

The graphical display enables the administrator to see the differences between the evaluation metrics such as Accuracy, F1-score, and ROC AUC. The graphical method serves as a more convenient way to compare models than anything based on only their numerical-values.

The model comparison module can come in handy in the development process as it provides support for choosing the best algorithm based on multiple evaluation metrics. The system enables comparison of various predictive performance indicators rather than purely accuracy.

5.1.9 Prediction Explainability Using SHAP

To improve the transparency of the prediction process, the system incorporates a SHAP-based Prediction Explanation Module.

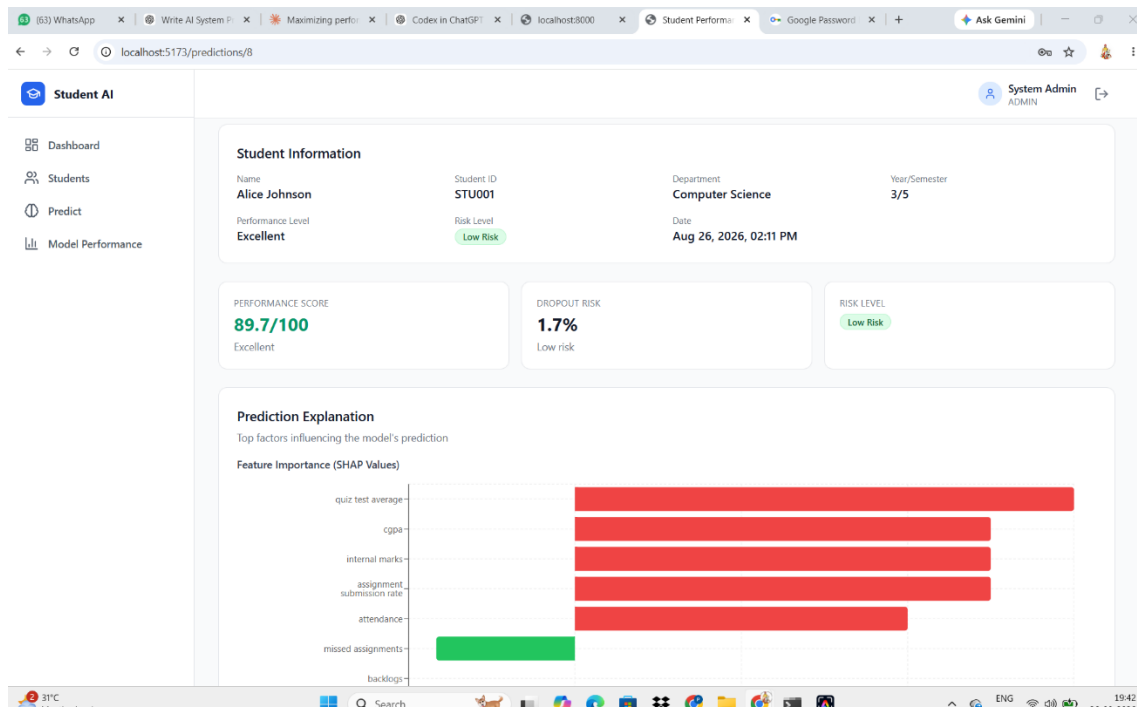


Figure 5.1.9: SHAP-Based Prediction Explanation

The interface of explainability shows the characteristics, which are essential in predicting the student's future performance. The demo that is made includes some features like Quiz Test Average, CGPA, Internal Marks, Assignment Submission Rate, Attendance, Missed Assignments, and Backlogs.

The values of each feature are presented on the horizontal axis in the form of a bar chart. The use of bar chart allows visualizing both negative and positive effects of the features on the prediction. In addition to this, the contributions of Positive Factors and Negative Factors are also distinguished in the interface.

In the application shown in the program, high academic indicators like Quiz Test Average, CGPA, Internal Marks are shown as main contributors while Missed Assignments have negative contribution. Among these features other indicators like attendance and assignment submission also have effects on the prediction. The reason this component of explainability is so crucial is that the system does not just provide the final performance classification. It allows the administrator to understand why a specific prediction was made. Knowing this helps educators understand the academic and behavioral aspects involved with an outcome.

Revised 5.2 Discussion

Unlike traditional academic assessment systems that primarily rely on the manual examination of grades and regular assessments, this new system combines student data

administration, machine learning mechanisms for forecasting, comparison of models, risk visualization, past-Predictions, and interpretation of results in one platform.

The screenshot of installation exhibits the working of the new system. The Student Management System enables organized accessing to data related to students about their education status while the administrative dashboard presents general information regarding the total number of students, distribution of risks as well as overall performance and attendance, and outcomes of predictions. The Recent Prediction section also shows the results along with the risks of producing these results, scores, the model of predictions, and the date of predictions.

The system can also compare many machine learning algorithms through March of models. This makes it possible to present the results of different sets of analyzes in the visual and tabular representations. According to the experimental results, the optimized Gradient Boosting model obtained 87.34% accuracy and had an F1-score of 87.42% for the testing dataset.

In addition, a nice feature of the implementation is the use of SHAP-based explainability. The Prediction Explanation Interface allows one to see the contributing features for each prediction along with its positive and negative impacts. As shown in the interface, academic factors like making quizzes, CGPA, internal marks, items submitted, and attendance are important; however, missed assignments and backlogs may negatively affect predictions. Nevertheless, the dataset used in this work does not have a dropout target to fulfil the requirements. Consequently, the validated results obtained are related only to the prediction of academic performance. The components of dropout-related and at-risk interfaces can be used in future for establishing a model that allows early predicting the exit probability; however, the model can only be claimed effective after training and evaluating the system with a relevant dataset providing dropout labels.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Conclusion

This paper outlines a machine learning framework for the analysis of students' academic performance and early identification of those who may need help with their academics. It tackles the shortcomings of existing academic assessment systems through the analysis of different student characteristics and use of machine learning algorithms to predict performance levels.

The designed system emphasizes the significance of several stages which include data preprocessing, feature extraction, model training, performance evaluation, prediction, and explainability.

The Data Preprocessing Module ensures that the student data is ready for use in machine learning algorithms through handling missing data values, discarding duplicates and inconsistent data entries, encoding of the categorical attributes, and converting the data to a required form. Classification

Models such as Logistic Regression, Decision Tree, Random Forest, KNN, SVM, and Gradient Boosting are tested to determine the best prediction technique.

The best performing optimized Gradient Boosting model was able to score an accuracy of 87.34%, precision of 87.95%, recall of 87.34%, and F1-Score of 87.42% on the test dataset. The use of SHAP-based explainability also aids in the understanding of the features driving the predictions by the model, where previous academic performance, especially that of G2, is found to have the most significant effect. Thus, this framework goes beyond academic score analysis to become an explainable and data-driven system for early identification.

6.2 Future Work

A number of enhancements can be made to the system in future releases to increase its functionality and usefulness for educational institutions. Most importantly, an enhancement would be to develop a separate prediction module for predicting the dropout risk using an appropriate labelled dataset having information about dropouts in the past. Such a labelled dataset would allow the machine learning models to classify the students based on their likelihood to drop out and perform an appropriate evaluation of dropout prediction performance.

In addition to that, a more powerful system can be designed by including various student information in real time like attendance data, assignment submission status, exam performance, activity in learning management system, and academic performance during semesters.

Another potential improvement is the creation of a dashboard for teachers and administrators by which predictions, risk scores, key contributing factors, and student progression can all be viewed graphically. The explainability component can also be improved by using SHAP individual explanations for each student, allowing the educator to know why the prediction was made.

The prediction algorithm can further be tested on larger and more varied datasets from other organizations to improve generalization. In future iterations, this model can also be extended through ensemble and deep learning techniques when enough data is present. Lastly, a recommendation and intervention component can be added in order to recommend the right support strategies for at-risk students, evolving the system into an early warning system with full functionality.

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